

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

May 2014

EC401 Data Communication and Networking

Date: 19/05/2014, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

- Q - 1 (a) Explain the following terms in brief: broadcasting and subnet. [02]
(b) What do you mean by topology? Compare mesh, bus, star and ring network topology in details. [05]
- Q - 2 (a) Describe the classification of various types of Transmission media. Explain guided media in brief. [07]
(b) Define multiplexing. Discuss types of multiplexing. [07]

OR

- Q - 2 (a) Explain digital signature in brief. [07]
(b) Define symmetric key cryptography and explain Data Encryption standard. [07]
- Q - 3 (a) Write a short note on Domain Name System. [07]
(b) Explain electronic mail in detail. [07]

OR

- Q - 3 (a) Explain LAN (Local Area Network) in detail. [05]
(b) Differentiate between connection oriented and connectionless services in brief. [02]
(c) Write short note on VOIP. [07]

SECTION – II

- Q - 4 (a) Define following terms (any two). [02]
A. Repeaters B. Hubs
C. Bridges D. Routers
- (b) Write a short note on IEEE 802.11. [05]
- Q - 5 (a) Explain TCP congestion control in detail [07]
(b) Explain CSMA/CA with necessary flow chart. [07]

OR

- Q - 5 (a) Explain GO-BACK N mechanism in case of (i) Delayed acknowledgment and (ii) Lost acknowledgment with necessary figures. [07]

- (b) What is Bluetooth protocol? Briefly explain Bluetooth architecture, frame structure and how communication is done between master and slave. [07]

Q - 6 (a) List out policies of open-loop congestion control and close loop congestion control in detail. [07]

- (b) Define switching. Compare circuit switching, message switching and packet switching in detail. [07]

OR

Q - 6 (a) Explain connection establishment in TCP using three way handshaking. [07]

- (b) Explain TCP segment header in detail. [07]
